

Draft

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1 Josephson Junction

1.1 Newton Update

The nonlinear system of equations to be solved is

$$\begin{aligned}
 -\alpha_{x1} E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}} + \beta_{x1} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - \gamma_{x1} E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} + \frac{1}{2} J_{J,x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} &= p_{x1} E_{x,i+\frac{1}{2},j,k}^n \\
 &+ \frac{1}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^n \right) \\
 &- \frac{1}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^n \right) \\
 &+ \frac{q_{x1}}{\Delta x \Delta y} \left(E_{y,i+1,j-\frac{1}{2},k}^n - E_{y,i,j-\frac{1}{2},k}^n \right) \\
 &- \frac{r_{x1}}{\Delta x \Delta y} \left(E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n \right) \\
 &- \frac{1}{2} J_{J,x,i+\frac{1}{2},j,k}^n,
 \end{aligned} \tag{1}$$

where

$$J_{J,x}^{n+\frac{1}{2}} = j_{c,x} \sin \left[\phi_x^n + \frac{\Delta t}{4} \eta_x \left(E_x^{n+\frac{1}{2}} + E_x^n \right) \right]. \tag{2}$$

Formulate into matrix form, along the y -line at fixed i, k , as

$$\mathbf{A}\mathbf{E} + \mathbf{J}(\mathbf{E}) = \mathbf{b}. \tag{3}$$

The unknown is the new electric field on that line,

$$E_j = E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}}. \tag{4}$$

$\mathbf{A}$ is the tridiagonal matrix (or low-rank perturbed tridiagonal matrix if a periodic boundary condition is used) formed from the linear ADI stencil, and $\mathbf{J}(\mathbf{E})$ is a diagonal matrix,

$$\mathbf{J}(\mathbf{E}) = \text{diag}(\mathbf{J}_1(\mathbf{E}), \dots, \mathbf{J}_M(\mathbf{E})), \tag{5}$$

where M is the number of unknowns on the line, and its j -th diagonal element is given by

$$\mathbf{J}_j(\mathbf{E}) = \frac{j_{c,x}}{2} \sin \left[\phi_{x,i+\frac{1}{2},j,k}^n + \frac{\Delta t}{4} \eta_x \left(E_j + E_{x,i+\frac{1}{2},j,k}^n \right) \right], \tag{6}$$

so $\mathbf{J}(\mathbf{E})$ changes only the diagonal. $\mathbf{b}$ is simply the right-hand side.

Define the equation to be solved as

$$\mathbf{F}(\mathbf{E}) = \mathbf{A}\mathbf{E} + \mathbf{J}(\mathbf{E}) - \mathbf{b} = \mathbf{0}. \tag{7}$$

The Jacobian matrix of the nonlinear term is

$$J_{\mathbf{J}}(\mathbf{E}) = \frac{\partial \mathbf{J}(\mathbf{E})}{\partial \mathbf{E}} = \text{diag}(d_1(\mathbf{E}), \dots, d_M(\mathbf{E})), \tag{8}$$

where

$$d_j(\mathbf{E}) = \frac{\partial \mathbf{J}_j(\mathbf{E})}{\partial E_j} = \frac{\Delta t}{8} \eta_x j_{c,x} \cos \left[\phi_{x,i+\frac{1}{2},j,k}^n + \frac{\Delta t}{4} \eta_x \left(E_j + E_{x,i+\frac{1}{2},j,k}^n \right) \right]. \tag{9}$$

Here η_x and $j_{c,x}$ are evaluated at edge $i + \frac{1}{2}, j, k$. The Jacobian of the full residual is therefore

$$J_{\mathbf{F}}(\mathbf{E}) = \frac{\partial \mathbf{F}(\mathbf{E})}{\partial \mathbf{E}} = \mathbf{A} + J_{\mathbf{J}}(\mathbf{E}), \quad (10)$$

which retains the tridiagonal structure. Thus, the Newton update is

$$\mathbf{E}^{(k+1)} = \mathbf{E}^{(k)} + \Delta \mathbf{E}^{(k)}, \quad (11)$$

where

$$\Delta \mathbf{E}^{(k)} = - \left(\mathbf{A} + J_{\mathbf{J}}(\mathbf{E}^{(k)}) \right)^{-1} \mathbf{F}(\mathbf{E}^{(k)}). \quad (12)$$

In practice, we solve the linear system

$$\left(\mathbf{A} + J_{\mathbf{J}}(\mathbf{E}^{(k)}) \right) \Delta \mathbf{E}^{(k)} = -\mathbf{F}(\mathbf{E}^{(k)}). \quad (13)$$